

Machine Learning

Model Selection & Assessment

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Lecture Overview

- 1** Lecture Overview
- 2** Model Selection
 - Validation Approach
 - Cross Validation Approach
 - PAC Approach
- 3** Model Assessment
- 4** Summary

Lecture Overview

By the end of this lecture you should:

- 1 Understand the concept of **Model Selection** in machine learning
- 2 Be aware of some practical approaches to model selection, including: **Validation techniques**, **Cross Validation techniques**, and **PAC learning**
- 3 Know the concept of **Model Assessment**

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Model Selection

- On several occasions we have seen the need to select various **hyperparameters** in order to fully define our learning algorithm (or our **model**). For example:
 - The ridge regression hyperparameter, λ
 - The degree of polynomial in polynomial regression
 - More abstractly: the 'complexity' measure in the bias-variance trade-off
- How we choose the best hyperparameter / learning algorithm / model for a particular problem is the task of **Model Selection**

Model Selection

- Note that model selection is primarily concerned with **ranking** a set of models on some basis, rather than estimating the **performance** of a particular model
- However, the way in which we perform a ranking is intimately connected with performance:

Model Selection

- Often our model selection approaches seek to approximate a measure of performance somehow
- And of course the measure of performance that we are ultimately interested in is the **Generalisation Loss**:

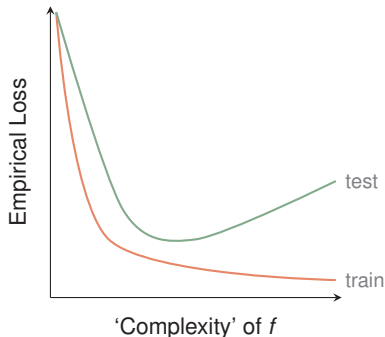
$$\mathbb{E}_{\mathcal{D}} [\mathcal{E}(f(\mathcal{X}), \mathcal{Y})]$$

- **Empirical Test Set Loss**, $\mathbb{E}_{\mathcal{S}_{\text{Test}}} [\mathcal{E}(f(\mathcal{X}), \mathcal{Y})]$, can act as a reasonable estimator of this quantity...
- ...But cannot be used for model selection!
- Can we use **Empirical Training Set Loss**, $\mathbb{E}_{\mathcal{S}_{\text{Train}}} [\mathcal{E}(f(\mathcal{X}), \mathcal{Y})]$ as an alternative?

Training Loss

■ No!

Recall our previous observation that a focus on such a **training loss** when selecting hyperparameters can lead us astray:



Training Loss

- So **Empirical Risk Minimisation** (ERM) (alone) is not enough to fit our hyperparameters / perform model selection...
- Instead we'll discuss three alternatives:
 - **Validation Techniques**
 - **Cross Validation Techniques**
 - **PAC Learning**

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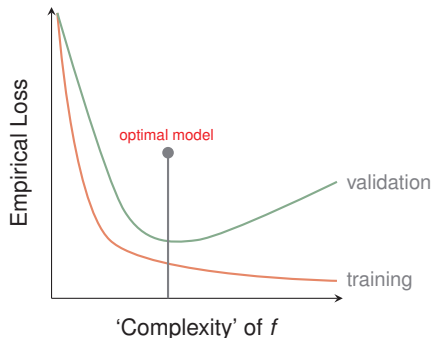
Validation Set Loss

- Here the basic idea is to split the initial training data into three sets: **Training**, **Validation**, and **Test** sets:
 - The first is used for **training** each possible model
 - The second is used to **select** the 'best' model
 - The third is used to **assess** the performance of that model



Validation Set Loss

- The motivation is that the **validation (set) loss** (the empirical loss on the validation set) will provide a proxy for the generalisation loss, which we then use to select an optimal model



Validation Set Bound

- We can prove that the validation loss is related to the generalisation loss, using **Hoeffding's Inequality**
- We will prove the result for the case of misclassification loss, using the following notation:

Notation

■ Validation Set:

$$\mathcal{S}_V = \{(\mathbf{x}^{(i)}, y^{(i)})\}_{i=1}^{n_V} \quad \text{where: } \mathcal{S}_V \sim \mathcal{D}^{n_V}$$

■ Loss Measure:

$$\mathcal{E}(f(\mathbf{x}), y) = \mathbb{I}[y \neq f(\mathbf{x})]$$

■ Generalisation Loss:

$$L(\mathcal{E}, \mathcal{D}, f) = \mathbb{E}_{\mathcal{D}} [\mathcal{E}(f(\mathcal{X}), \mathcal{Y})]$$

■ Validation Loss:

$$L(\mathcal{E}, \mathcal{S}_V, f) = \frac{1}{n_V} \sum_{i=1}^{n_V} \mathcal{E}(f(\mathbf{x}^{(i)}), y^{(i)})$$

Validation Set Bound

■ Let us recall **Hoeffding's Inequality**:

Let $Z_1, \dots, Z_n$ be independent random variables, such that each Z_i is bounded by the interval $[a_i, b_i]$, then for any $\epsilon > 0$:

$$\mathbb{P}(\mathbb{E}[\bar{Z}] - \bar{Z} \geq \epsilon) \leq \exp\left(-\frac{2n^2\epsilon^2}{\sum_{i=1}^n (b_i - a_i)^2}\right)$$

Validation Set Bound

- Apply this to i.i.d. $\mathcal{Z}_i$, such that $\mathcal{Z}_i = \mathbb{I}[\mathcal{Y}^{(i)} \neq f(\mathcal{X}^{(i)})]$, means that $(b_i - a_i) = 1$ and:

$$\mathbb{P} \left(\mathbb{E}_{\mathcal{D}} [\mathbb{I}[\mathcal{Y} \neq f(\mathcal{X})]] \geq \frac{1}{n_{\mathcal{V}}} \sum_{i=1}^{n_{\mathcal{V}}} \mathbb{I}[\mathcal{Y}^{(i)} \neq f(\mathbf{x}^{(i)})] + \epsilon \right) \leq e^{-2n_{\mathcal{V}}\epsilon^2}$$

- And so:

$$\mathbb{P} (\mathbb{E}_{\mathcal{D}} [\mathbb{I}[\mathcal{Y} \neq f(\mathcal{X})]] \geq \mathbb{E}_{\mathcal{S}_{\mathcal{V}}} [\mathbb{I}[\mathcal{Y} \neq f(\mathcal{X})]] + \epsilon) \leq e^{-2n_{\mathcal{V}}\epsilon^2}$$

Validation Set Bound

- If we wish the probability of the occurrence of this event to be no greater than some quantity $\delta \in [0, 1]$ then:

$$\delta = e^{-2n_V \epsilon^2}$$
$$\implies \epsilon = \sqrt{\frac{\ln(1/\delta)}{2n_V}}$$

- This setting of ϵ will ensure that the probability of the occurrence of the complement of this event will be greater than $(1 - \delta)$:

$$\mathbb{P} \left(\mathbb{E}_{\mathcal{D}} [\mathbb{I}[y \neq f(\mathbf{x})]] < \mathbb{E}_{S_V} [\mathbb{I}[y \neq f(\mathbf{x})]] + \sqrt{\frac{\ln(1/\delta)}{2n_V}} \right) > (1 - \delta)$$

Validation Set Bound

- Equivalently, we can state the **Validation Set Bound**: Let f be some predictor function, then with probability of at least $(1 - \delta)$ over the choice of validation set, $\mathcal{S}_V$:

$$L(\mathcal{E}, \mathcal{D}, f) \leq L(\mathcal{E}, \mathcal{S}_V, f) + \sqrt{\frac{\ln(1/\delta)}{2n_V}}$$

Validation Set Bound

- This holds for a single function f , but we wish to use the validation loss to choose from amongst a set of functions
- Using the **Union Bound** we can prove that a similar bound holds for all members of a finite hypothesis class, $\mathcal{F}$:

$$\mathcal{F} = \{f_1, \dots, f_r\} \quad \text{of size:} \quad |\mathcal{F}|$$

Validation Set Bound

- Recall that by the Union Bound, for events A and B , the following inequality holds:

$$\begin{aligned}\mathbb{P}(A \vee B) &\leq \mathbb{P}(A) + \mathbb{P}(B) \\ \implies 1 - \mathbb{P}(A \vee B) &\geq 1 - (\mathbb{P}(A) + \mathbb{P}(B)) \\ \implies \mathbb{P}(\neg(A \vee B)) &\geq 1 - (\mathbb{P}(A) + \mathbb{P}(B)) \\ \implies \mathbb{P}(\neg A \wedge \neg B) &\geq 1 - (\mathbb{P}(A) + \mathbb{P}(B)) \quad \text{by DeMorgan}\end{aligned}$$

Validation Set Bound

- Let A be the event, (for some function f^A):

$$L(\mathcal{E}, \mathcal{D}, f^A) > L(\mathcal{E}, \mathcal{S}_v, f^A) + \sqrt{\frac{\ln(1/\delta')}{2n_v}}$$

- Then by the validation set bound:

$$\mathbb{P}(A) < \delta'$$

- Similarly let B be the event, (for some function f^B):

$$L(\mathcal{E}, \mathcal{D}, f^B) > L(\mathcal{E}, \mathcal{S}_v, f^B) + \sqrt{\frac{\ln(1/\delta')}{2n_v}}$$

- Then by the validation set bound:

$$\mathbb{P}(B) < \delta'$$

Validation Set Bound

- Combining these, for all $f \in \{f^A, f^B\}$:

$$\mathbb{P}\left(L(\mathcal{E}, \mathcal{D}, f) \leq L(\mathcal{E}, \mathcal{S}_V, f) + \sqrt{\frac{\ln(1/\delta')}{2n_V}}\right) \geq 1 - 2\delta'$$

- Then, if we set $\delta = 2\delta'$ then, for all $f \in \{f^A, f^B\}$:

$$\mathbb{P}\left(L(\mathcal{E}, \mathcal{D}, f) \leq L(\mathcal{E}, \mathcal{S}_V, f) + \sqrt{\frac{\ln(2 \times 1/\delta)}{2n_V}}\right) \geq 1 - \delta$$

- Our result for all $f \in \{\mathcal{F}\}$ where the number of functions contained in $\mathcal{F}$ is $|\mathcal{F}|$ follows in a similar way.

Validation Set Bound

- With probability of at least $(1 - \delta)$ over the choice of $\mathcal{S}_V$:

$$\forall f \in \mathcal{F} \quad L(\mathcal{E}, \mathcal{D}, f) \leq L(\mathcal{E}, \mathcal{S}_V, f) + \sqrt{\frac{\ln(|\mathcal{F}|/\delta)}{2n_V}}$$

Validation Set Bound

- In other words, the validation loss is a good guide to the generalisation loss provided that:
 - $\mathcal{F}$ is not too large (i.e. we don't try too many models)
 - n_V is not too small
- If either of these conditions is not true, then the validation and generalisation losses will drift apart as we begin to **overfit** the model / hyperparameters
- For example the Kaggle leader board phenomenon:
 - Entrants routinely top the validation leader board, but perform poorly on the final test board

Stratification

- One way of alleviating the small sample size problem is to **stratify** our sampling
- **Stratification** involves sampling the data for each set (training/validation/test) such that each contains the same proportion of data points with particular combinations of features
- This helps to ensure that training, validation, and test sets are all representative of $\mathcal{D}$

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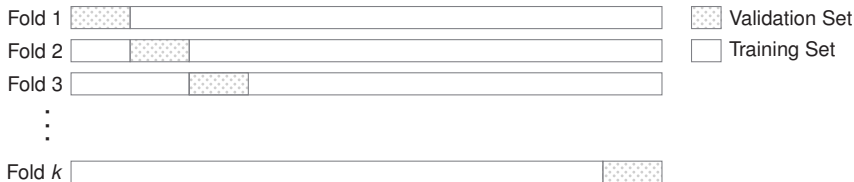
Cross Validation

- But data is often not plentiful at all
- If we wish to retain most of our data for training then we must adopt a different approach:
- **Cross Validation** allows us to perform validation, while still maintaining our training data set

Cross Validation

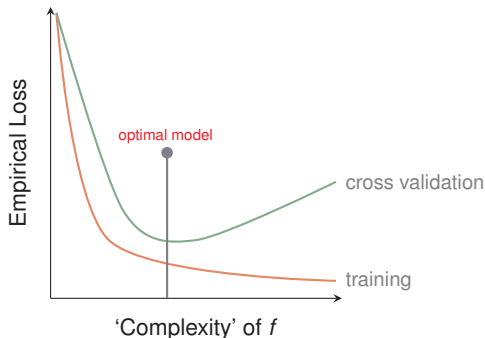
- Here the basic idea is to split the initial training data into k sets (or **folds**)
- The learning algorithm is run k times, for each model, each time using all the folds but one as a **training** set, $\mathcal{S} \setminus \mathcal{S}_k$, and the remaining fold as a **validation** set, $\mathcal{S}_k$
- The validation performance, for a particular model f , is averaged across all k folds to give the **cross validation loss**:

$$L_{CV_k}(\mathcal{E}, \mathcal{S}, f) = \frac{1}{k} \sum_{i=1}^k L_{\mathcal{S} \setminus \mathcal{S}_k}(\mathcal{E}, \mathcal{S} \setminus \mathcal{S}_k, f)$$



Cross Validation

- Again, the motivation is that the **cross validation loss** will provide a proxy for the generalisation loss, which we then use to select an optimal model



Cross Validation: k selection

- What value should we select for k ?
- $k = n$ is known as **Leave One Out (LOO)** cross validation
 - It maximises the training set for each fold iteration - and so reduces bias in the hypothesis learned for each such iteration
 - But it is computationally expensive
 - And each validation fails to give a good estimate of the generalisation error associated with the learned hypothesis
- Empirical analysis suggests that $k = 5$ or 10 is better
 - This is used in practice almost as standard
 - And is computationally cheaper

Cross Validation: Comments

- Cross Validation works well in practice...
- ...But a rigorous understanding of why is an open question
- And in certain circumstances it must be treated with care, for example:
 - **Time Series Data:**
 - Only **sequential** cross validation should be performed
 - Randomising would artificially enhance validation set performance
 - Neighbouring points are likely to be similar and should therefore be kept together

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The PAC Approach

- Here we begin with the generalisation loss...
- ...And seek to formulate a **probabilistic** 'worst-case' **bound** on this quantity in terms of:
 - The observable **empirical training loss**
 - Some **complexity penalty**, which takes into account the size of the **representation space**, $\mathcal{F}$
- Then we perform model selection by searching for the model / hyperparameter choice which gives rise to the **tightest bound**

PAC Learning

- The learner receives samples and must select a hypothesis, f , from, $\mathcal{F}$, such that with high probability (**probably**) the selected function will have low generalisation loss (will be **approximately correct**)
- This must be true for any probability of success, $(1 - \delta)$, any approximation loss target, ϵ , and any data generating process, $\mathcal{D}$

PAC Learning: Procedure

- These **PAC generalisation bounds** can be generated using the **Uniform Convergence** approach.
This is a two step procedure:
 - First we seek some finite sample **concentration inequality** that is independent of $\mathcal{D}$. We apply this to one function, f
 - Second we seek to apply this inequality **uniformly** across all f in $\mathcal{F}$ via the **Union Bound** or some measure of the functional complexity of $\mathcal{F}$
- Let's take a look at this approach in a few different **classification** settings:

Notation

■ Training Set:

$$\mathcal{S} = \{(\mathbf{x}^{(i)}, y^{(i)})\}_{i=1}^n \quad \text{where: } \mathcal{S} \sim \mathcal{D}^n$$

■ Loss Measure:

$$\mathcal{E}(f(\mathbf{x}), y) = \mathbb{I}[y \neq f(\mathbf{x})]$$

■ Generalisation Loss:

$$\mathbb{E}_{\mathcal{D}} [\mathcal{E}(f(\mathcal{X}), \mathcal{Y})]$$

■ Training Loss:

$$\mathbb{E}_{\mathcal{S}} [\mathcal{E}(f(\mathcal{X}), \mathcal{Y})]$$

PAC Bound: Finite $\mathcal{F}$, Consistent Learner

- A **consistent learner** is a learning algorithm which only outputs hypotheses, $f \in \mathcal{F}$, that are consistent with the training data, (i.e. $\mathbb{E}_{\mathcal{S}}[\mathcal{E}(f(\mathcal{X}), \mathcal{Y})] = 0$)

- Assuming $(0 < \epsilon < 1)$ and $(0 < \delta < 1)$:

$$\mathbb{P}(\text{a particular } f \in \mathcal{F}, \text{ with } \mathbb{E}_{\mathcal{D}}[\mathcal{E}(f(\mathcal{X}), \mathcal{Y})] > \epsilon, \\ \text{is consistent with 1 training ex.}) < (1 - \epsilon) < e^{-\epsilon}$$

- So:

$$\mathbb{P}(\text{a particular } f \in \mathcal{F}, \text{ with } \mathbb{E}_{\mathcal{D}}[\mathcal{E}(f(\mathcal{X}), \mathcal{Y})] > \epsilon, \\ \text{is consistent with } n \text{ training ex.}) < (1 - \epsilon)^n < e^{-\epsilon n}$$

PAC Bound: Finite $\mathcal{F}$, Consistent Learner

- For this to apply to all $f \in \mathcal{F}$ simultaneously, we begin by applying the **Union Bound** to obtain:

$$\mathbb{P}(\mathbb{E}_{\mathcal{D}}[\mathcal{E}(f(\mathcal{X}), \mathcal{Y})] > \epsilon, \text{ for at least one } f \in \mathcal{F}) < |\mathcal{F}|e^{-\epsilon n}$$

- **Theorem:**

For all $(0 < \epsilon < 1)$, $(0 < \delta < 1)$, all data generating distributions, $\mathcal{D}$, then with probability of at least $(1 - \delta)$:

$$\forall f \in \mathcal{F} \quad \mathbb{E}_{\mathcal{D}}[\mathcal{E}(f(\mathcal{X}), \mathcal{Y})] \leq \frac{1}{n} \ln \left(\frac{|\mathcal{F}|}{\delta} \right)$$

- But what if we wish to access a *more realistic* representation, where our learners can make mistakes?

PAC Bound: Finite $\mathcal{F}$, Agnostic Learner

- An **agnostic learner** is a learning algorithm which is not restricted in the hypotheses it can learn
- Let us recall **Hoeffding's Inequality**:
Let $\mathcal{Z}_1, \dots, \mathcal{Z}_n$ be independent random variables, such that each $\mathcal{Z}_i$ is bounded by the interval $[a_i, b_i]$, then for any $\epsilon > 0$:

$$\mathbb{P}(\mathbb{E}[\bar{\mathcal{Z}}] - \bar{\mathcal{Z}} \geq \epsilon) \leq \exp\left(-\frac{2n^2\epsilon^2}{\sum_{i=1}^n (b_i - a_i)^2}\right)$$

PAC Bound: Finite $\mathcal{F}$, Agnostic Learner

- Apply this to i.i.d. $\mathcal{Z}_i$, such that $\mathcal{Z}_i = \mathbb{I}[y^{(i)} \neq f(\mathbf{x}^{(i)})]$, means that $(b_i - a_i) = 1$ and:

$$\mathbb{P} \left(\mathbb{E}_{\mathcal{D}} [\mathbb{I}[y \neq f(\mathbf{x})]] \geq \frac{1}{n} \sum_{i=1}^n \mathbb{I}[y^{(i)} \neq f(\mathbf{x}^{(i)})] + \epsilon \right) \leq e^{-2n\epsilon^2}$$

- And so:

$$\mathbb{P} (\mathbb{E}_{\mathcal{D}} [\mathbb{I}[y \neq f(\mathbf{x})]] \geq \mathbb{E}_{\mathcal{S}} [\mathbb{I}[y \neq f(\mathbf{x})]] + \epsilon) \leq e^{-2n\epsilon^2}$$

PAC Bound: Finite $\mathcal{F}$, Agnostic Learner

- For this to hold for all $f \in \mathcal{F}$ simultaneously, we begin by applying the **Union Bound** to obtain:

$$\mathbb{P}(\mathbb{E}_{\mathcal{D}}[\mathbb{I}[\mathcal{Y} \neq f(\mathcal{X})]] \geq \mathbb{E}_{\mathcal{S}}[\mathbb{I}[\mathcal{Y} \neq f(\mathcal{X})]] + \epsilon, \text{ for at least one } f \in \mathcal{F}) \leq |\mathcal{F}|e^{-2n\epsilon^2}$$

- **Theorem:**

For all $(0 < \epsilon < 1)$, $(0 < \delta < 1)$, all data generating distributions, $\mathcal{D}$, then with probability of at least $(1 - \delta)$:

$$\forall f \in \mathcal{F} \quad \mathbb{E}_{\mathcal{D}}[\mathbb{I}[\mathcal{Y} \neq f(\mathcal{X})]] \leq \mathbb{E}_{\mathcal{S}}[\mathbb{I}[\mathcal{Y} \neq f(\mathcal{X})]] + \sqrt{\frac{1}{2n} \ln \left(\frac{|\mathcal{F}|}{\delta} \right)}$$

- But what if we wish to access a *larger* representation?

PAC Bound: Infinite $\mathcal{F}$, Agnostic Learner

- For infinitely large function classes the previous bound becomes trivial because $|\mathcal{F}| \rightarrow \infty$
- We need a more subtle way of measuring the **size**, or **functional capacity**, or **complexity** of $\mathcal{F}$
- There are a number of different possible measures, for example:
 - **Rademacher Complexity**
 - **Covering Numbers**
 - **PAC-Bayes**
- Using such a measure we can develop a similar PAC bound to the ones which we've already seen:

PAC Bound: Infinite $\mathcal{F}$, Agnostic Learner

- For example for **misclassification loss**, and the **Rademacher complexity** measure we can derive:

- **Theorem:**

For all $(0 < \epsilon < 1)$, $(0 < \delta < 1)$, all data generating distributions, $\mathcal{D}$, then with probability of at least $(1 - \delta)$:

$$\forall f \in \mathcal{F} \quad \mathbb{E}_{\mathcal{D}} [\mathbb{I}[y \neq f(\mathcal{X})]] \leq \mathbb{E}_{\mathcal{S}} [\mathbb{I}[y \neq f(\mathcal{X})]] + 2\mathbb{R}(\mathcal{F} \circ \mathcal{S}) + 4\sqrt{\frac{2}{n} \ln \left(\frac{4}{\delta} \right)} \quad (1)$$

- Here $\mathbb{R}(\mathcal{F} \circ \mathcal{S})$ denotes the Rademacher complexity of $\mathcal{F} \circ \mathcal{S} = \{f(\mathbf{x}^{(i)}) | f \in \mathcal{F}\}_{i=1}^n$

PAC Bound: Infinite $\mathcal{F}$, Agnostic Learner

- Characterising $\mathbb{R}(\mathcal{F} \circ \mathcal{S})$ is:
 - Beyond the scope of this course!
 - Depends on the training data and on $\mathcal{F}$
 - For some interesting representations a finite quantity...
 - ...even if $\mathcal{F}$ contains an infinite number of functions!

PAC Bound: Model Selection

- So Theorem (1) gives us a worst case upper bound in probability for the generalisation loss for a wide variety of models
- We can calculate this bound for different models ($\mathcal{F}$) characterised by, for example, different hyperparameter settings
- We can perform model selection by picking the model which gives rise to the tightest of these bounds

PAC Bound: Comments

- The PAC approach has advantages:
 - It is not limited to misclassification loss, it can be applied to a variety of other loss functions
 - It allows models to be compared with training data alone, while avoiding overfitting
- However the bounds are often loose:
 - Thus they are often poor estimators of the generalisation loss itself
 - For model selection, PAC bound optimisation tends to underperform cross validation
- But the value of this approach is much broader:
 - It provides a principled basis for understanding why learning algorithms should work
 - And the process of bound formation then optimisation offers a method for principled novel algorithm creation...as we'll see...

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Model Assessment

- Having chosen a final model, the task of **model assessment** is to estimate the generalisation loss of the model on new data
- As we already saw, the most desirable approach, is to observe model performance on a **validation set**, and then to form a worst case **probabilistic bound** on the generalisation loss
- But in the absence of sufficient data to attempt this, what should we do?

Model Assessment

- The **PAC Bounds** are too loose to be useful for assessment
- The **Cross Validation** loss is a practical and empirically good alternative estimator
- ...While not theoretically well understood, it is widely used in practice

Model Assessment: Metrics

- A related issue is what our metric of efficacy should be
- Learning algorithms are typically designed for canonical loss metrics, e.g.:
 - Mean square error
 - Misclassification loss
- But often we're interested in something different, e.g.:
 - Profitability
 - Quality of Life

Model Assessment: Metrics

- We could design a new algorithm to focus on bespoke metrics
 - But this is hard

- Or we could use our metric of interest in the cross validation phase
 - This can be effective...
 - ...But is a heuristic halfway house which allows us to aim an inadequate algorithm at the right objective

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Summary

- 1 Both **Model Selection** and **Model Assessment** involve attempts at characterising the generalisation loss
- 2 The **Validation Set** approach is a desirable way of tackling these tasks, but requires abundant data
- 3 The **PAC** approach is a well-motivated alternative but is often practically deficient
- 4 The **Cross Validation** approach is a widely used (though poorly understood) practical alternative